Discrete Mathematics (MATH 208)
Fall 2022
Friday, 11/04/2022
Homework 10: Chapters 25, 26, 27

Score: / 10

Name (Print): ____________________

Chapter 25 - Linear Diophantine Equations

$\boxed{1 \text{ pts}}$ 1. Find all integer solutions to $14x + 77y = 69$.

$\boxed{1 \text{ pts}}$ 2. Find all integer solutions to $14x + 77y = 70$.

$\boxed{1 \text{ pts}}$ 3. If you all you have are dimes and quarters, in how many ways can you pay a \$7 bill? (For example, one way would be 10 dimes and 24 quarters.)

Chapter 26 - Modular Arithmetic

$\boxed{\text{1 pts}}$ 1. The marks on a combination lock are numbered 0 to 39. If the lock is at mark 19, and the dial is turned one mark clockwise, it will be at mark 18. If the lock is at mark 19 and turned 137 marks clockwise, at what mark will it be?

$\boxed{\text{1 pts}}$ 2. Solve: $3x \equiv 5 \pmod{8}$.

$\boxed{\text{1 pts}}$ 3. Solve: $12x \equiv 9 \pmod{88}$.

$\boxed{\text{1 pts}}$ 4. Solve: $33x \equiv 183 \pmod{753}$.

Chapter 27 - Integers in Other Bases

1 pts

1. Using $A, B, \cdots, F$ to represent base 16 digits from 10 to 15 (if needed), convert the decimal integer 3177 to the following bases:
   (a) Base 2

   (b) Base 6

   (c) Base 16

2 pts

2. Fill in the following *base* 8 addition and multiplication tables:

+	0	1	2	3	4	5	6	7
0								
1								
2								
3								
4								
5								
6								
7								

$\times$	0	1	2	3	4	5	6	7
0								
1								
2								
3								
4								
5								
6								
7								